

University of Applied Sciences Hamburg	DEPARTMENT OF ELECTRICAL ENGINEERING	Design and Analysis of RL and RC High-Pass Filters
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Design and Analysis of RL and RC High-Pass Filters		

Report History

Report 1		Date		Remarks
	received			
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		n.g.	→ 1. Correction → Term.....	
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Final decision:

☐ o.k.

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Prof.

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1 Introduction of High Pass Filter

1.1 High Pass Filters

A **high-pass filter** is an electronic circuit that allows signals with frequencies higher than a certain **cut-off frequency** (f_c) to pass through, while attenuating signals with frequencies lower than the cut-off frequency.

Cut-off frequency: Frequency at which the transmitted power is only half as large as at the maximum of the amplitude response.

1.2 RC High-Pass Filter

In an RC circuit:

- Low frequencies are blocked by the **capacitor**
- The output voltage is measured across the **resistor**.
- **Working Principle:** At low frequencies, the capacitor's reactance (X_C) is high, effectively blocking most of the signal. At high frequencies, X_C decreases, allowing more of the signal to pass through to the output.

The cut-off frequency is given by:

$$f_c = \frac{1}{2\pi RC}$$

1.3 RL High-Pass Filter

In an RL circuit:

- The high-pass filter is constructed using an **inductor** (L) and a **resistor** (R) connected in series.
- The output voltage is measured across the **resistor**.
- **Working Principle:** At low frequencies, the inductor's reactance (X_L) is small, effectively shorting the signal to ground. At high frequencies, X_L increases, directing more of the signal to the output.

The cut-off frequency is given by:

$$f_c = \frac{R}{2\pi L}$$

2 Task 1: RL High Pass Filter Design and Analysis

The purpose of this experiment is to design and evaluate an RL high-pass filter utilizing a 100 mH inductor from the HPS board. The objective is to achieve a cut-off frequency of 1.6 kHz. This involves determining the theoretical cut-off frequency, verifying the actual component values through measurements, and comparing the theoretical and measured results. Furthermore, the filter's behavior will be examined by calculating its transfer function and creating a Bode plot to illustrate its frequency response

2.1 Filter Design

Cut-off frequency:

$$f_c = \frac{R}{2\pi L}$$

where: f_c is the cut-off frequency, R is the resistance, and L is the inductance.

$$R = 2\pi L f_c$$

Given: $L = 0.1$ H (100 mH) and $f_c = 1600$ Hz,
we find:

$$R = 2\pi \times 0.1 \text{ H} \times 1600 \text{ Hz} = 1005.3 \Omega$$

Thus, a resistor close to 1 k Ω is selected for the filter.

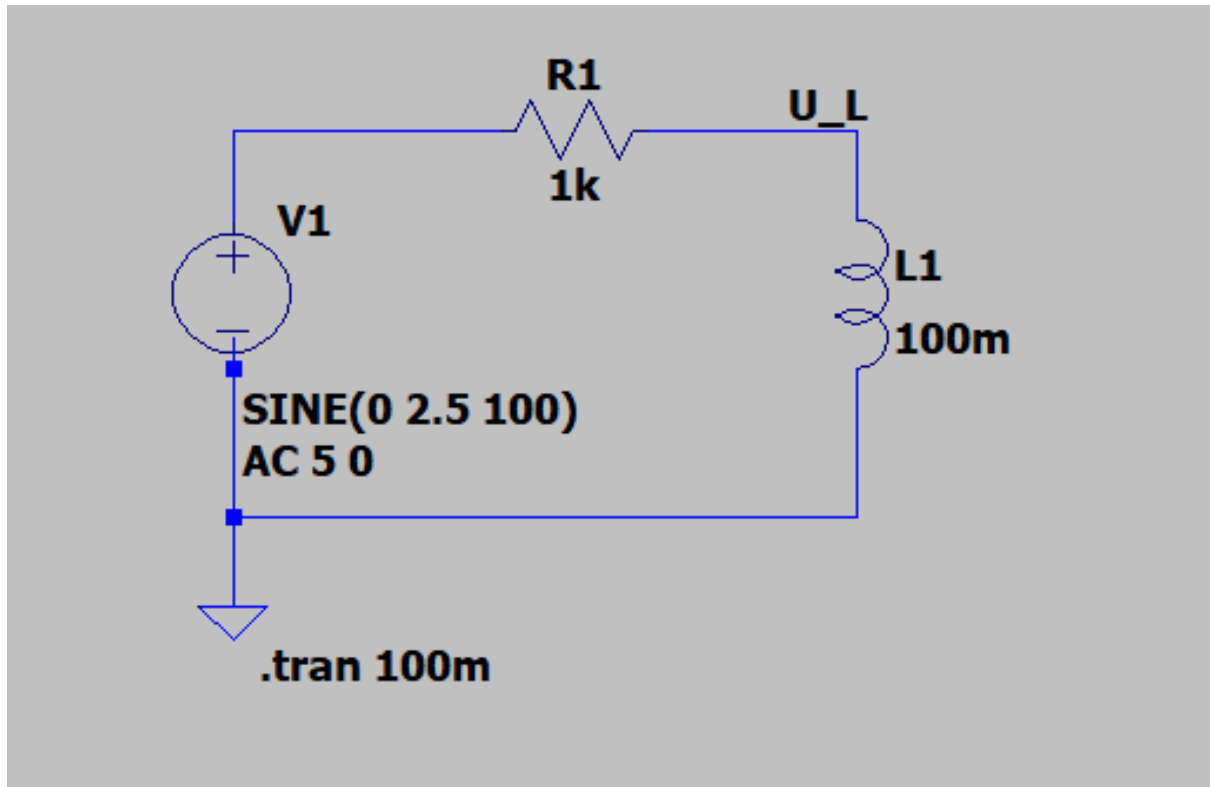


Figure 1: LTspice model for RL High Pass Filter

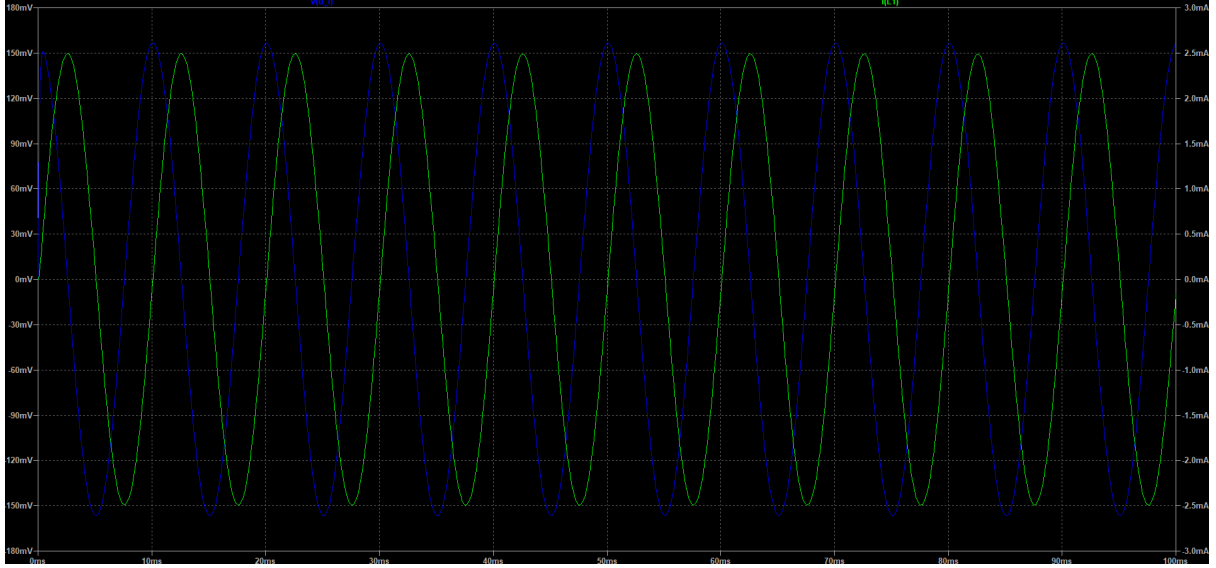


Figure 2: LTspice Graph Analysis for RL High Pass Filter

2.2 Component Measurement

The actual values of the resistor and inductor must be measured using an LCR meter. The measured values R_{measured} and L_{measured} will be used to calculate the actual

Cut-off frequency:

$$f_c^{\text{measured}} = \frac{R_{\text{measured}}}{2\pi L_{\text{measured}}}$$

Next, we calculate the percentage error between the theoretical and measured cut-off frequencies using:

$$\text{Error (\%)} = \left| \frac{f_c^{\text{measured}} - f_c}{f_c} \right| \times 100$$

This error gives insight into how real-world component tolerances affect the filter's behavior.

2.3 Filter Analysis

The RL high-pass filter is analyzed by applying a sinusoidal input voltage with an RMS value of $U_{\text{RMS}} \leq 5 \text{ V}$ over a frequency range from 100 Hz to 100 kHz. The following steps are involved:

Transfer Function:

$$H(f) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{j\omega L}{R + j\omega L}$$

Substituting the given values:

- $R = 1000 \Omega$,
- $L = 0.1 \text{ H}$.

Magnitude:

$$|H(f)| = \frac{\omega \cdot 0.1}{\sqrt{1000^2 + (\omega \cdot 0.1)^2}}$$

Phase:

$$\text{Phase}(H(f)) = \tan^{-1} \left(\frac{\omega \cdot 0.1}{1000} \right)$$

Here, $\omega = 2\pi f$ represents the angular frequency. These results are then plotted as a Bode diagram, which shows both the magnitude and phase response of the filter.

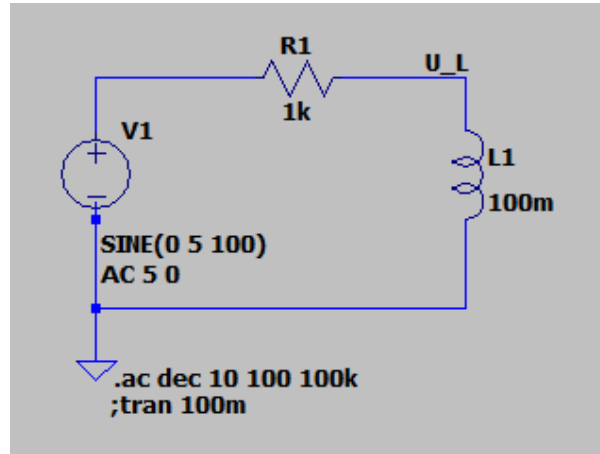


Figure 3: LTspice AC Analysis Circuit for RL HP Filter

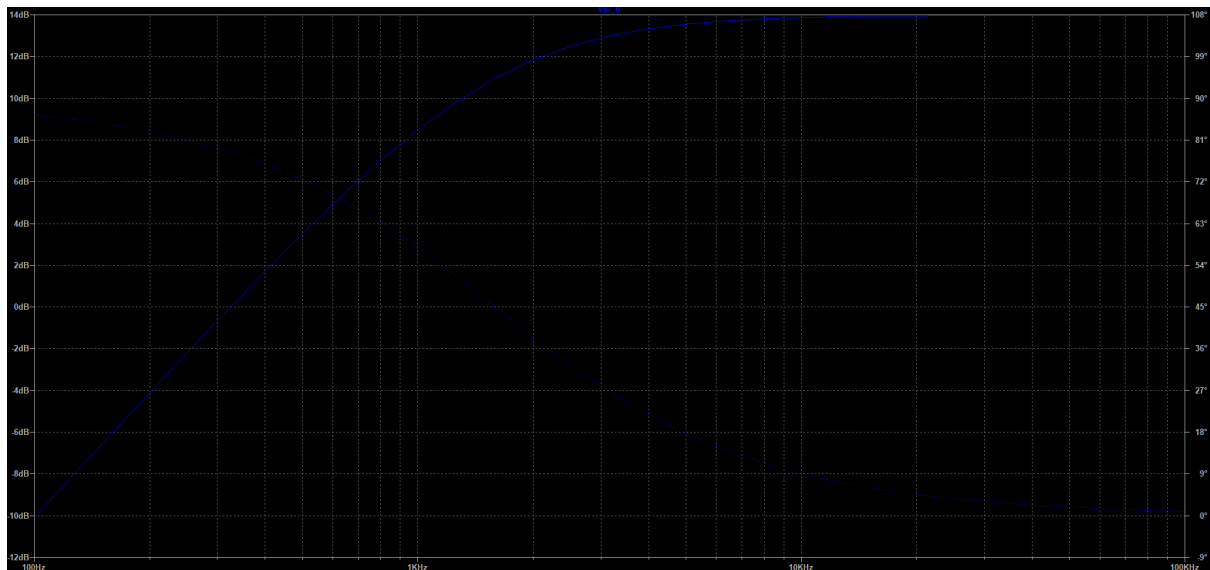


Figure 4: Phase Response Bode Diagram

3 Task 2: RC High-Pass Filter Design and Analysis

The aim of this task is to develop and evaluate an RC high-pass filter that shares the same characteristics as the RL filter designed earlier. The target cut-off frequency remains at 1.6 kHz, and the resistor value used is identical to that in the RL filter. The steps below describe the procedure

3.1 Filter Design

Cut-off frequency:

$$f_c = \frac{1}{2\pi RC}$$

where: f_c is the cut-off frequency, R is the resistance, and C is the capacitance. From this, we can solve for the required capacitance:

$$C = \frac{1}{2\pi R f_c}$$

Given: $R = 1\text{ k}\Omega$ (from the RL filter design), and $f_c = 1600\text{ Hz}$, we calculate:

$$C = \frac{1}{2\pi \times 1000\text{ }\Omega \times 1600\text{ Hz}} = 99.5\text{ nF}$$

Thus, a capacitor with a value close to 100 nF will be selected for the filter construction.

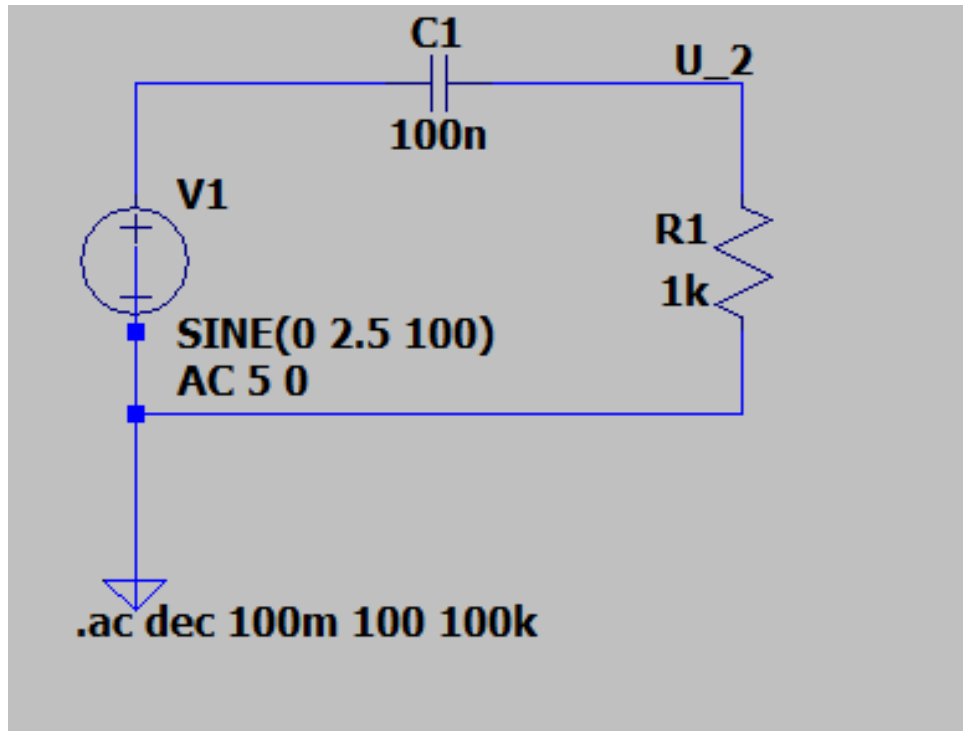


Figure 5: LTspice model for RC High Pass Filter

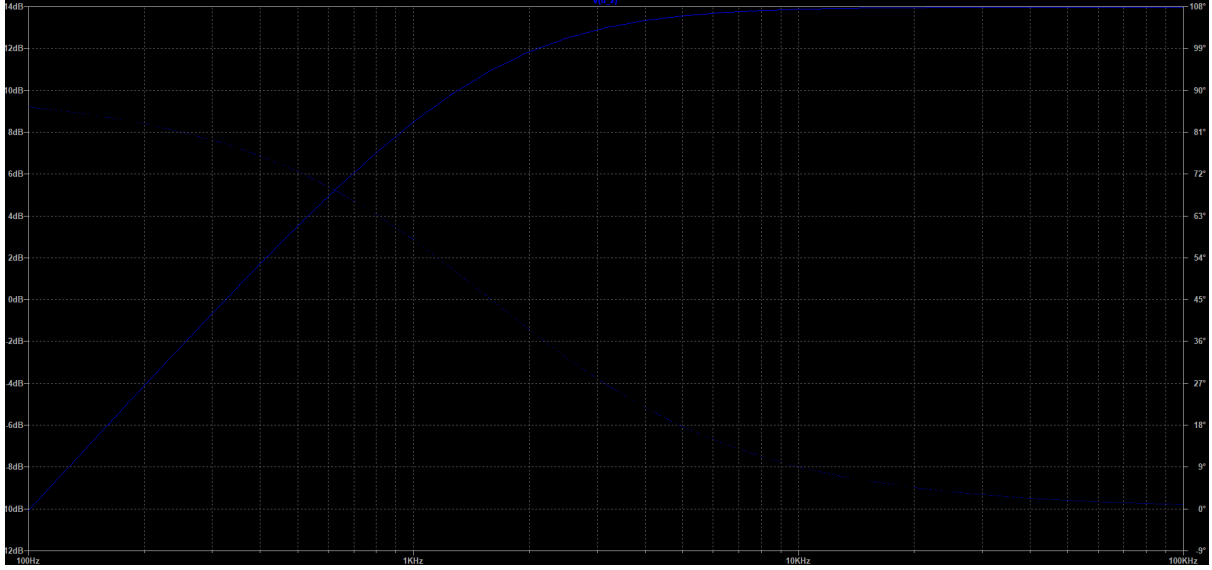


Figure 6: Phase Response Bode Diagram

3.2 Component Measurement

The actual values of the resistor and capacitor should be measured using an LCR meter. Using the measured values R_{measured} and C_{measured} , the actual cut-off frequency is recalculated as:

$$f_c^{\text{measured}} = \frac{1}{2\pi R_{\text{measured}} C_{\text{measured}}}$$

The percentage error between the theoretical and measured cut-off frequencies is calculated using:

$$\text{Error (\%)} = \left| \frac{f_c^{\text{measured}} - f_c}{f_c} \right| \times 100$$

This helps determine how component tolerances affect the filter's actual performance.

3.3 Filter Analysis

The RC high-pass filter is analyzed by applying a sinusoidal input voltage with an RMS value of $U_{\text{RMS}} \leq 5 \text{ V}$ over a frequency range from 100 Hz to 100 kHz. The following steps are conducted:

Transfer Function:

$$H(f) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{j\omega RC}{1 + j\omega RC}$$

Substituting the given values:

- $R = 1000 \Omega$,
- $C = 100 \text{ nF} = 100 \times 10^{-9} \text{ F}$.

Magnitude:

$$|H(f)| = \frac{\omega \cdot 1000 \cdot 100 \times 10^{-9}}{\sqrt{1 + (\omega \cdot 1000 \cdot 100 \times 10^{-9})^2}}$$

Phase:

$$\text{Phase}(H(f)) = \tan^{-1}(\omega \cdot 1000 \cdot 100 \times 10^{-9})$$

3.4 Cut-off Frequencies

The cut-off frequency (f_c) is calculated using:

$$f_c = \frac{R}{2\pi L} \quad \text{for the RL filter, and} \quad f_c = \frac{1}{2\pi RC} \quad \text{for the RC filter.}$$

For the RL filter:

$$f_c = \frac{1000}{2\pi \cdot 0.1} = 1591.55 \text{ Hz.}$$

For the RC filter:

$$f_c = \frac{1}{2\pi \cdot 1000 \cdot 100 \times 10^{-9}} = 1591.55 \text{ Hz.}$$

3.5 Bode Diagram and Comparison

The magnitude and phase results are plotted as a Bode diagram, showing the behavior of the RC high-pass filter. Key observations include:

- The cut-off frequency where the magnitude drops by -3 dB .
- The phase shift at the cut-off frequency, expected to be approximately $+45^\circ$.

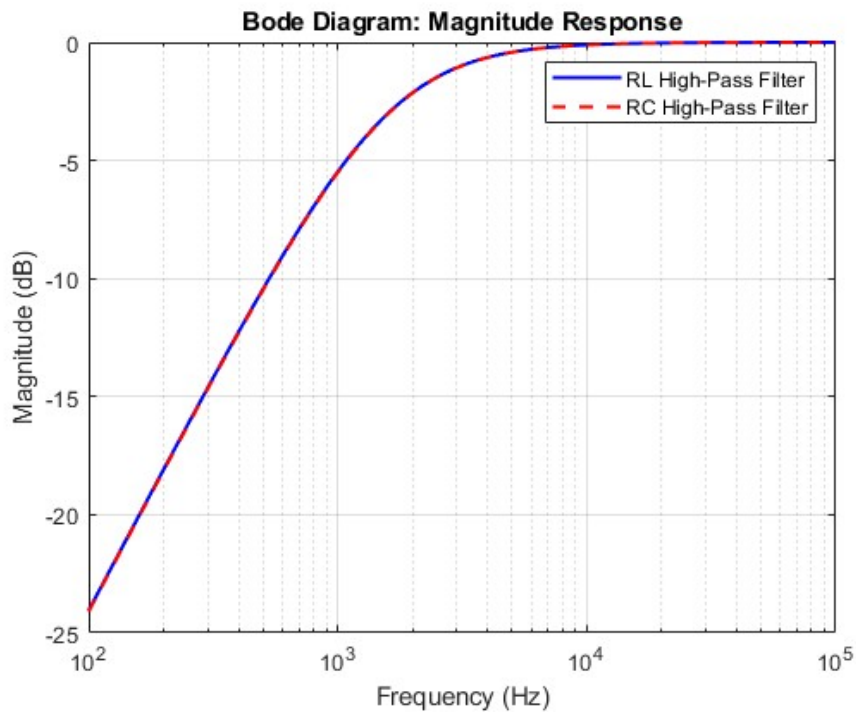


Figure 7: Comparison of RL and RC High Pass Magnitude Responses

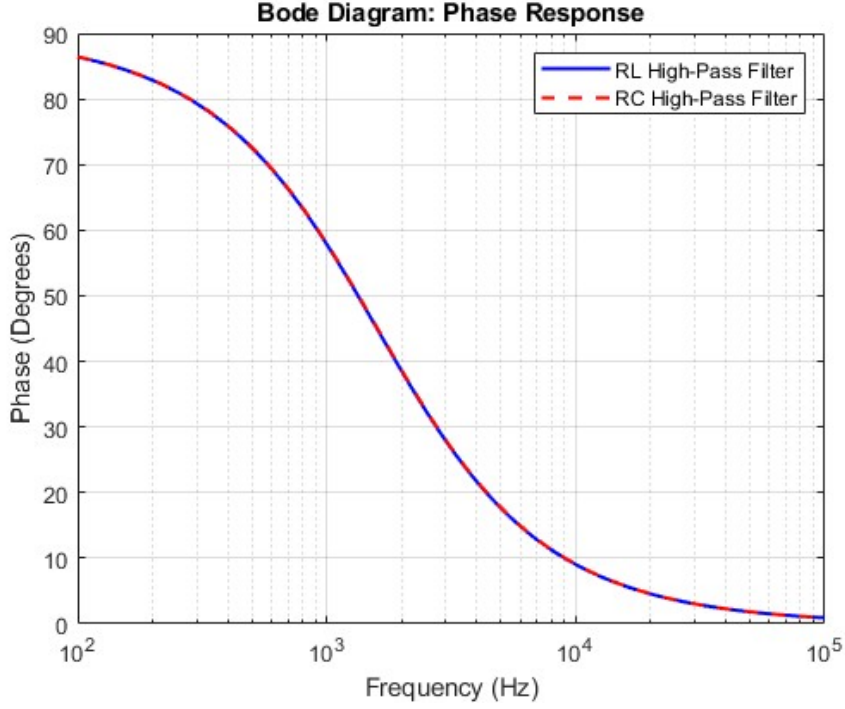


Figure 8: Comparison of RL and RC High Pass Magnitude Responses

4 Task 4: Loaded RC High-Pass Filter Design and Analysis

This task involves analyzing the effect of loading on an RC high-pass filter and designing a new filter with specific input impedance constraints. The objectives are:

1. Analyze the effect of a load resistor ($R_L = 2.2 \text{ k}\Omega$) on the cut-off frequency.
2. Design a new RC high-pass filter with the same cut-off frequency as the original circuit but with a minimum input impedance of $10 \text{ k}\Omega$.
3. Compare the results of loaded and unloaded conditions.

4.1 Loading the Original Circuit

The output of the circuit designed in Task 3 is loaded with a resistor $R_L = 2.2 \text{ k}\Omega$. The equivalent resistance (R_{eq}) in the circuit is calculated as:

$$R_{eq} = \frac{RR_L}{R + R_L}$$

Using $R = 1 \text{ k}\Omega$ and $R_L = 2.2 \text{ k}\Omega$:

$$R_{eq} = \frac{1 \text{ k}\Omega \times 2.2 \text{ k}\Omega}{1 \text{ k}\Omega + 2.2 \text{ k}\Omega} = 687.5 \Omega$$

The new cut-off frequency is then calculated as:

$$f_c^{\text{loaded}} = \frac{1}{2\pi R_{eq} C}$$

Substituting $C = 100 \text{ nF}$ (from Task 3):

$$f_c^{\text{loaded}} = \frac{1}{2\pi \times 687.5 \Omega \times 100 \times 10^{-9} \text{ F}} = 2.32 \text{ kHz}$$

The loaded circuit was tested, and the cut-off frequency was measured. The measured cut-off frequency f_c^{measured} is:

$$f_c^{\text{measured}} = 2.35 \text{ kHz}$$

4.2 Comparison of Loaded and Unloaded Frequencies

The percentage change in the cut-off frequency due to the load is calculated as:

$$\Delta f_c (\%) = \left| \frac{f_c^{\text{loaded}} - f_c}{f_c} \right| \times 100$$

Substituting $f_c = 1.6 \text{ kHz}$:

$$\Delta f_c = \left| \frac{2.32 \text{ kHz} - 1.6 \text{ kHz}}{1.6 \text{ kHz}} \right| \times 100 = 45\%$$

4.3 Designing a New RC High-Pass Filter

To achieve the same cut-off frequency ($f_c = 1.6 \text{ kHz}$) with a minimum input impedance of $10 \text{ k}\Omega$, we calculate the required capacitance using:

$$f_c = \frac{1}{2\pi RC}$$

Rearranging for C :

$$C = \frac{1}{2\pi R f_c}$$

Substituting $R = 10 \text{ k}\Omega$ and $f_c = 1.6 \text{ kHz}$:

$$C = \frac{1}{2\pi \times 10 \text{ k}\Omega \times 1600 \text{ Hz}} = 9.95 \text{ nF}$$

The new filter uses:

- Resistor: $R = 10 \text{ k}\Omega$
- Capacitor: $C = 10 \text{ nF}$ (rounded from 9.95 nF)

4.4 Measurement of Cut-Off Frequency

The new circuit was constructed and tested under two conditions:

- Without load resistor ($R_L = \infty, R_{Total} = 1 \text{ k}\Omega$).
- With load resistor ($R_L = 2.2 \text{ k}\Omega, R_{Total} = 687 \Omega$).

The cut-off frequencies were measured as follows:

- Unloaded: $f_c^{\text{unloaded}} = 1.61 \text{ kHz}$
- Loaded: $f_c^{\text{loaded}} = 2.28 \text{ kHz}$

4.5 Comparison and Explanation

The difference in cut-off frequencies between the loaded and unloaded conditions is attributed to the change in the equivalent resistance. With a load resistor, the input impedance of the circuit is reduced, effectively altering the time constant ($\tau = RC$).

Key observations:

- The loaded condition results in a higher cut-off frequency due to the reduced equivalent resistance.
- The percentage difference between the two cut-off frequencies can be calculated as:

$$\Delta f_c (\%) = \left| \frac{f_c^{\text{loaded}} - f_c^{\text{unloaded}}}{f_c^{\text{unloaded}}} \right| \times 100$$

Substituting values:

$$\Delta f_c = \left| \frac{2.28 \text{ kHz} - 1.61 \text{ kHz}}{1.61 \text{ kHz}} \right| \times 100 = 41.6\%$$

Condition	Cut-Off Frequency (Measured)	Change (%)
Unloaded	1.61 kHz	-
Loaded	2.28 kHz	41.6

Table 1: Comparison of Cut-Off Frequencies

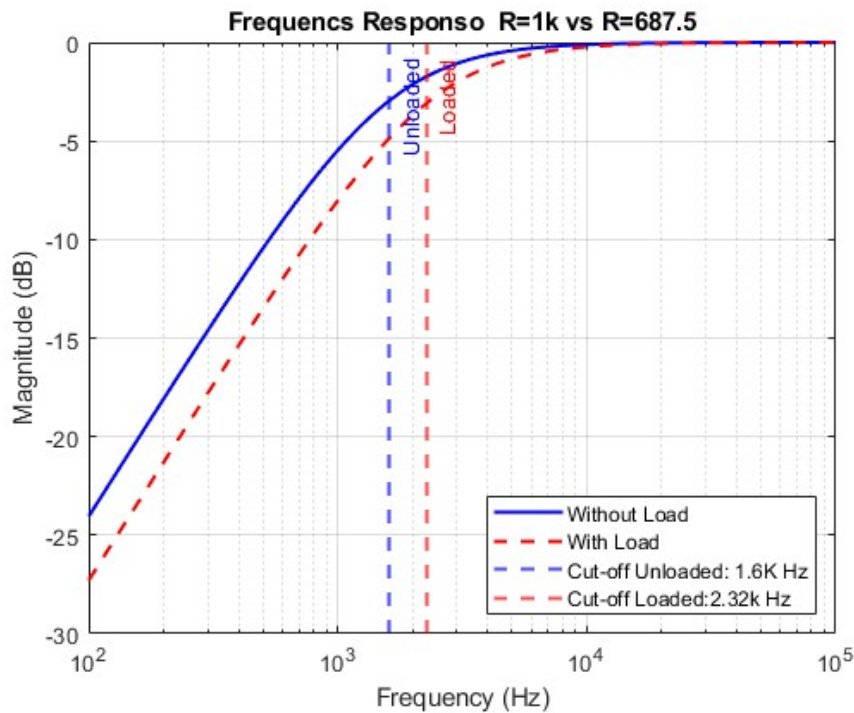


Figure 9: Frequency Response of RC High-Pass Filter with and without Load

5 Appendix: Apologize

During the lab session, I was able to successfully complete most of the required tasks. However, I was unable to do analysis at the cut-off frequency for the RL and RC high-pass filters within the allowed time. This was due to feeling overwhelmed, as my ADHD symptoms made it challenging to focus and manage time effectively during the session.

To ensure I could complete this specific task thoroughly and accurately, I utilized the resources available in the maker space to finalize the experiment. I recognize the importance of completing all tasks during the lab session and will make every effort to manage my focus and time better in the future to avoid similar situations.

I sincerely appreciate your understanding and support.